Data Prep
manifest.csv ->
train/val/test
split
-> label mapping

Dataset construction

Training pipeline
chunk into 4s
windows
(loop-fill
remainder)
-> log-mel ->
cache ->
normalize

Val/Test pipeline
full natural-
length clip
-> log-mel ->
cache ->
normalize

DataLoaders
train: batch=32,
shuffled
val/test:
batch=1

Training Loop (CNNBaseline, 20 epochs)

Optimize
Adam +
CrossEntropyLoss

Validate each
epoch
track loss /
accuracy /
macro-F1

Adapt LR
ReduceLROnPlateau
on macro-F1

Checkpoint best
model
by highest
val_macro_f1

Evaluation &
Analytics
[cnn_analytics.py]
classification
report, confusion
matrix,
training curves,
sample browser,
duration-bias
check, UMAP/PCA,
GradCAM